

```
30.
31. # Return distribution
32. plt.subplot(3, 2, 2)
33. plt.hist(data['Returns'].dropna(), bins=50)
34. plt.axvline(0, color='red', linestyle='--')
35. plt.title('Return Distribution')
36. plt.grid(True)
37.
38. # Volatility over time
39. plt.subplot(3, 2, 3)
40. plt.plot(data['Volatility'])
41. plt.title('21-Day Rolling Volatility (Annualized %)')
42. plt.grid(True)
43.
44. # Autocorrelation of returns
45. plt.subplot(3, 2, 4)
46. plot_acf(data['Returns'].dropna(), lags=30, ax=plt.gca())
47. plt.title('Autocorrelation of Returns')
48.
49. # Cumulative returns
50. plt.subplot(3, 2, 5)
51. cum_returns = (1 + data['Returns']).cumprod()
52. plt.plot(cum_returns)
53. plt.title('Cumulative Returns')
54. plt.grid(True)
55.
56. # Q-Q plot to check for normality
57. plt.subplot(3, 2, 6)
58. from scipy import stats
59. stats.probplot(data['Returns'].dropna(), plot=plt)
60. plt.title('Q-Q Plot of Returns')
61.
62. plt.tight_layout()
63. plt.show()
64.
65. # 3. Stationarity test
66. result = adfuller(data['Returns'].dropna())
67. print(f"\nADF Statistic: {result[0]:.4f}")
```